Sarah Henia



AI Engineer

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Portfolio | LinkedIn | GitHub

SUMMARY

AI engineer who designs and ships agentic systems that run in production: LLM pipelines where a human approves every write, graph memory and hybrid retrieval over real company data, and the data pipelines and cloud infrastructure under them on AWS and Azure, built with governance in mind.

EXPERIENCE

The SamurAI — *Remote, USA*
Software Architect & AI Solution Integrator — *Feb 2026 - Present*

- Designed and built **Dawn**, an autonomous **AI agent** for project management, live in production from March 2026: turns recorded meetings into tasks with owner, priority and due date, with **human-in-the-loop** approval before any write to Microsoft Planner.
- Engineered its **graph memory**: a SurrealDB **knowledge graph** with **hybrid RAG** (vector search plus recency boost) and a **Redis context cache (CAG)**, integrated with the **Microsoft Graph API**.
- Designed its **governance and security layer**: identity gate, per-agent policies, HMAC-signed agent messages, audit log, confidence gating, and sensitive data routed to a local Llama model.
- Built **multi-model routing** through OpenRouter with automatic fallback; deployed on hardened **AWS EC2** with **GitHub Actions CI/CD** and **observability** in Logfire and Langfuse.
- Designing and building a **multi-agent market-intelligence system** for a Middle East cybersecurity client: bilingual (English/Arabic) **data ingestion**, classification and schema-enforced **LLM extraction** into a **SurrealDB knowledge graph**, with human review before any CRM write.
- Designing its **AI governance**: an **AI gateway** for every model and tool call (credential, budget, trace), controls mapped to **NIST AI RMF** and **ISO/IEC 42001**, and **evaluation metrics** on a labelled set run on every change.

Product Manager — *Aug - Dec 2025*

- Managed the roadmap and backlog of **Dojo**, an AI-powered platform that generates on-demand test labs, turning stakeholder needs into specifications and sprint priorities.

CED Tunisia — *Tunis, Tunisia*
Software Engineering Intern — *Feb - Aug 2026*

- Built **Collaboris**, a real-time user-presence library for enterprise web apps, shipped as two installable packages: an **Angular 19** library (npm) and an **ASP.NET Core 8** SDK (NuGet) over **SignalR** and **Redis**. Delivered in three releases under department-head review.
- Built **PresenceAI**, the data and AI layer: **change-data capture** streaming live Redis signals into a Cosmos DB journal (**OLTP to OLAP**, Python/FastAPI), queried in natural language through a governed **Azure OpenAI** assistant over **MCP**.
- Implemented **GDPR compliance by design** (data minimisation, consent, 30-day TTL retention, fail-closed access, cost caps) and covered the AI layer with **unit tests**.
- Deployed the full stack to **Azure** (App Service, SQL Server, Azure Cache for Redis) through **Azure DevOps** build and release pipelines, migrating the API's data layer from PostgreSQL to **SQL Server**.

EDUCATION

National Diploma of Engineer, Computer Science · ESPRIT, Tunis — *Sep 2023 - Sep 2026*
Bachelor's Degree, Business Computing (E-Business), with Business Intelligence coursework · ESSECT, Tunis — *Sep 2020 - Jun 2023*

SKILLS

AI: Agentic and multi-agent systems, LLMs, RAG, CAG, graph memory, knowledge graphs, MCP, AI gateway, LLM evaluation, human-in-the-loop, LangChain, Azure OpenAI, OpenRouter

Data: Data modeling, ETL, change-data capture, OLTP/OLAP, medallion architecture; SurrealDB, PostgreSQL, SQL Server, MongoDB, Redis, Cosmos DB, Neo4j

Build: Python, FastAPI, C#, ASP.NET Core, Node.js, TypeScript, Angular, React, REST, GraphQL, WebSockets, Microsoft Graph API

Cloud & Quality: AWS, Azure, Docker, GitHub Actions, CI/CD, Linux, observability (Logfire, Langfuse), pytest; GDPR, NIST AI RMF, ISO/IEC 42001

CERTIFICATIONS

Cisco CCNA (2024) · AWS Certified Cloud Practitioner (2025) · NVIDIA DLI, 4 courses incl. Building RAG Agents with LLMs and Adversarial ML (2025)

LANGUAGES

Arabic (native) · English (C1) · French (B2)